Control The World With Bluetooth LE

Get A Headstart On This Amazing Technology

***Bluetooth* Low Energy** is a wireless radio standard that promises low power, low latency, low range, and low bandwidth communication.

Component Cost

Sensor Quality

Mobile Computing Availability

Examples

Examples

Decisions, Decisions ...

Bluetooth: Classic vs LE

	Classic (2.1+EDR)	Low Energy (4.0)
Latency	30 - 100ms	2.5ms
Power	~ 200mA, Tx	~ 10mA, Tx
Range	100m	50m
Throughput	~ 1 Mbit/s	~ 0.3 Mbit/s
# of Slaves	7	Undefined
Channels	79, 1-MHz Wide	40, 2-MHz Wide
Audio	Yes	No
Profiles	Rigid	Open
MFi	Required	Not Required
iOS Frameworks	EAAccessory	CoreBluetooth

Advertising and ...

Scanner

Broadcaster

... Connecting

Connection

Master
(Central, Client)

Slave
(Peripheral, Server)

Profiles

Heart Rate Profile

Profiles -> Services

Name	UUID
Heart Rate Service	0x180D
Device Information Service	0x180A
Battery Service	0x180F

Profiles -> Services -> Characteristics

Name	UUID	Service
Heart Rate Measurement	0x2A37	0x180D
Body Sensor Location	0x2A38	0x180D
Manufacturer Name	0x2A29	0x180A
Model Number	0x2A24	0x180A
Battery Level	0x2A19	0x180F

Coming Up ...

9:41 AM
Messages
Calendar
Photos
Camera
Videos
Maps
Weather
Passbook
Notes
Reminders
Clock
Stocks
Newsstand
iTunes
App Store
Game Center
Settings
Phone
Mail
Safari
Music

To Contact Me ...

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